

Formula Sheet for Chapters 5&6

Future Value of Lump Sum	$FV_n = PV(1 + i)^n$
Present Value of Lump Sum	$PV = \frac{FV_n}{(1+i)^n}$
FV of an Ordinary Annuity	$FVA_N = PMT \left[\frac{(1+i)^N - 1}{i} \right]$
FV of an Annuity Due	$FV_{due_N} = PMT \left[\frac{(1+i)^N - 1}{i} \right] \times (1+i)$
PV of an Ordinary Annuity	$PVA_N = PMT \left[\frac{1 - \frac{1}{(1+i)^N}}{i} \right]$
PV of an Annuity Due	$PVA_N = PMT + PMT \left[\frac{1 - \frac{1}{(1+i)^N}}{i} \right]$
PV of an Ordinary Per	$PV_{\text{Perpetuity Ordinary}} = \frac{PMT}{i}$
PV of a Perpetuity Due	$PV_{\text{Perpetuity Due}} = PMT + \frac{PMT}{i}$
PV of an Uneven Cash Flow	$PV = \frac{CF_1}{(1+i)^1} + \frac{CF_2}{(1+i)^2} + \frac{CF_3}{(1+i)^3} + \dots + \frac{CF_N}{(1+i)^N} = \sum_{t=1}^N \frac{CF_t}{(1+i)^t}$
Periodic interest rate	$i_{\text{Periodic}} = \frac{\text{Stated Annual Rate}}{\text{Number of Payments per year}}$
Effective Annual Rate (EAR or EFF%)	$EAR = \left[1 + \frac{i}{M} \right]^M - 1$
Amortized Loan Calculations: Step-1: Balance for year 0: <i>Amount of loan</i> Step-2: Calculate the amount of payments. Step-3: Interest for year t is: $\text{Balance}_{t-1} \times i$ Step-4: Principal Payment for year t is: $\text{Payment} - \text{Interest}_t$ Step-5: Balance for year t is: $\text{Balance}_{t-1} - \text{Principal}_t$	
- Quoted (nominal) interest rate (r) for corporate bond = $r^* + IP + DRP + LP + MRP$ - Quoted (nominal) interest rate (r) for T-bill & T-bond = $r^* + IP + MRP$	

